

Position Paper: An Emoji-Based Interactive App to Aid Emotion Expression for Autistic Children

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ABSTRACT

Autism spectrum disorder (ASD) is a neurodevelopmental condition typically characterized by two core characteristics as restricted, repetitive behaviors and impairment in social communication. As such, a large portion of technology studies within the Human-Computer Interaction (HCI) community has been focused on improving social skills for autistic individuals in the past decade. In the meantime, autism is primarily perceived as a medical deficit that requires "corrections." Technology often requires autistic children to adjust their behaviors and traits to cater to "normative" modes of interaction that are deemed to be appropriate for the neurotypical society. The goal of most technology research for autistic individuals is to satisfy the extrinsic purpose while ignoring autistic children are the actual primary users using the technology. An ample number of studies have been evaluated on using different approaches to educate autistic children on comprehending and interpreting the emotions of neurotypical individuals. Yet, few works have taken the opposite approach that invites the neurotypical user to read the emotions of autistic children during social interaction. In this work, we propose an emoji-based interactive app that aims to aid autistic children in expressing their emotions while encouraging neurotypical individuals to comprehend the emotions of autistic children to foster bi-directional social interaction.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools; Ubiquitous and mobile computing design and evaluation methods.**

KEYWORDS

Autism, Children, Ubiquitous Computing, Emotion, Interaction Design, Personalized Training

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1 INTRODUCTION

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition typically characterized by the impairment in social communication and restricted, repetitive sensory-motor behaviors [14]. In the past decade, the rising trend of technology works has involved assisting autistic individuals especially children with autism, among the field of Human-Computer Interaction (HCI) [21]. Considering the deficit in social communication as one of the two core characteristics of autism, a large portion of these studies has been placed on improving social skills for autistic individuals. In the meantime, autism is prominently perceived as a medical deficit that requires "corrections." As a result, autistic individuals are often "forced" to adjust their behaviors and traits to cater to "normative" modes of social interaction deemed to be appropriate for the neurotypical community, such as learning communication engagement [17, 20, 22] and imitating others [4, 6, 8]. The goal of a large number of technology research for autistic individuals "reverts" to satisfy the extrinsic purpose of the neurotypical community, while ignoring autistic children as the actual primary users using the technology.

In addition, an ample number of studies have been evaluated on using different approaches to teach autistic children to comprehend and interpret the emotions of neurotypical individuals, to list a few: [2, 12, 13]. Yet, few works have taken the opposite approach that considers the neurotypical user to read the emotions of the autistic user during social interaction, which motivates our work. In this study, we propose an emoji-based interactive app that aims to aid autistic children in expressing their emotions while inviting

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neurotypical individuals to comprehend the emotions of autistic children to foster bi-directional social interaction.

2 BACKGROUND

2.1 Autism

Autism Spectrum Disorder (ASD) is a developmental disorder that often has difficulties in social interaction and communication with restrictive and repetitive behaviors [14]. Early intervention towards autistic children has presented compelling results [3, 9], including some of our work: [23, 25]. Studies on technology-based intervention for autistic individuals have popularly integrated the Social Communication, Emotional Regulation and Transactional Support (SCERTS) model [18], which consists of three main components: building competence in social communication, emotional regulation, and transactional support.

Autism is also predominantly perceived as a medical deficit that requires "corrections," leading to technology-based studies that often require autistic children to adjust their behaviors and traits to cater to "normative" modes of interaction that are deemed to be appropriate for the neurotypical society though autistic children are supposed to be in the position of the primary user of these technologies [21]. A significant shift towards the understanding of autism has occurred in recent years among the autism research community, which considers autism more as neurodiversity [10] rather than disability. Neurodiversity defines autism as one of the neurological variations while further encouraging self-advocacy and self-empowerment for autistic individuals towards technologies [21]. In the meantime, the study [21] has revealed that the technology study tailed for the diverse interests, needs, and desires of autistic children remains largely unexplored.

2.2 Assistive Technology for Social Skills

Given that the deficit in social communication is a core characteristic of autism, a significant number of studies has been focused on improving social skills for autistic individuals using multitudes of technologies on diverse aspects, such as learning communication engagement [17, 20, 22], imitating [4, 6, 8] and emotion recognition [2, 12, 13]. Porcino et al. [17] developed mobile games that aim to improve the learning and communication skills of autistic children under the supervision of therapists. Their results address the importance of the involvement of the therapists and the necessity of an elastic application interface that supports high customization and personalization. Palestra et al. [15] propose a humanoid robot that is designed to act as a social mediator to educate autistic children on eye contact, joint attention, symbolic play, and emotion recognition. The result was reported to be promising, with a significant increase in eye contact and imitation behaviors. The study in [19] engages autistic children in a virtual interaction environment of the Minecraft community while allowing them to explore freely. The improved social engagement was reported. More recently, Li et al. [13] designed an augmented reality (AR) based virtual agent social game to teach children with autism basic facial expressions and improve their emotional and communication skills.

An increasing number of studies have been evaluating different approaches to educate autistic children on comprehending

and interpreting social interaction modes of neurotypical individuals [21], such as learning how to engage more with communication, present empathy and recognize others' emotions. Most works "conveniently" place autistic individuals in the position of secondary users to cater to the extrinsic purpose defined by the neurotypical society, though they are supposed to be the beneficiaries of these technologies. Few works have taken the opposite approach that elicits neurotypical users to read and respond to the social interaction modes of autistic individuals (such as their emotions) during social interaction to better engage in bi-directional social interaction, which essentially motivates our proposed work.

2.3 Challenges

Though emojis have been widely accepted and applied among social networks to represent different emotional states [5], little work has been done to investigate the existence of such linkage among autistic individuals, especially autistic children. Early practice [1] has shown that autistic children sometimes find it challenging to interpret complex emotions, such as surprise and fear, while they find it relatively easier to distinguish emotions from more basic emotions. A more comprehensive empirical study will be needed to investigate the intrinsic causes of this sort of misinterpretation/mismatch, or autistic individuals might naturally have their own preference over the mapping pairs between emotion and emojis. The methodology employed in [11] for the Deaf and Hard of Hearing (DHH) community represents a potentially promising approach for future reference.

In the meantime, Tigwell and Flatla [24] have reported inconsistency in emoji representation between different mobile platforms (such as Android and iOS). Thus, during the design stage, applying a consistent set of emoji representations across platforms is essential to reduce the potential misinterpretation, which could otherwise discourage the self-advocacy of autistic children.

Furthermore, considering the diverse traits of autistic children, the study has suggested involving autistic children in the design process with the benefits of tailoring technologies to their needs and desires and further making the technology more suitable and acceptable for autistic children [21]. The downside part of this approach places on over-concentrated details driven by recent experiences of autistic children, which makes it challenging to convert it into actual design [7]. As such, involving strategy, such as mindful interpretation, to unfold autistic children's thinking becomes essential [16].

In addition, the proposed system aims to foster bi-directional social interaction. Evaluating the extent to which both parties would appreciate the system is challenging and should be carefully designed, as well as the system's effectiveness in encouraging interaction.

3 CONCLUSION

In this position paper, instead of fully placing the liability of social communication on autistic individuals, we propose an emoji-based interactive app that could aid autistic children in expressing their emotions and feelings while eliciting neurotypical individuals to comprehend the emotions of autistic children to foster the bi-directional social interaction. In the meantime, we hope this paper

could further inspire the autism research community to devote work to endorse self-advocacy and self-empowerment for autistic individuals towards technologies while considering placing autistic individuals in the position of first user during the design.

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